



CEFEPIME 1g INJECTION

COMPOSITION:

Each vial contains
Cefepime Hydrochloride buffered with L-Arginine
equivalent to Cefepime 1g.

PRESENTATION:

White to pale yellow sterile powder that provides a
clear solution after reconstitution.

INDICATIONS:

Adult :

Vaxcel Cefepime 1g Injection is indicated in adults
for the treatment of the following infections when
caused by susceptible strains of bacteria :
Lower respiratory tract infections (including
pneumonia and bronchitis).
Urinary tract infections (both complicated, including
pyelonephritis and uncomplicated infections).
Skin and skin structure infections.
Intra-abdominal infections (including peritonitis and
biliary tract infections).
Gynecologic infections.
Septicemia.
Empiric treatment in febrile neutropenic patients.

Paediatrics :

Vaxcel Cefepime 1g Injection is indicated in
pediatric patients for treatment of infections when
caused by susceptible bacteria :
Pneumonia
Urinary tract infections (both complicated and
uncomplicated, including pyelonephritis)
Skin and skin structure infections
Septicemia
Empiric treatment in febrile neutropenia.
Culture and susceptibility studies should be
performed when appropriate to determine
susceptibility of the causative organism(s) to
cefepime. Empiric therapy with Vaxcel Cefepime
may be instituted before results of susceptibility
studies are known; however, once these results
become available, the antibiotic treatment should be
adjusted accordingly. Because of its broad spectrum
of bactericidal activity against gram-positive and
gram-negative bacteria, Vaxcel Cefepime can be
used as monotherapy prior to identification of the
causative organism(s). In patients who are at risk
of mixed aerobic-anaerobic infection, particularly if
bacteria not susceptible to cefepime may be present
(see MICROBIOLOGY), concurrent initial therapy
with anti-anaerobic agent is recommended before
the causative agents may or may not be necessary,
depending on the susceptibility profile.

PHARMACODYNAMICS:

Cefepime is a forth-generation cephalosporin
antibiotic. It is a bactericidal agent that acts by

inhibition of bacterial cell wall synthesis. It has a broad spectrum
activity against a wide range of gram-positive and gram-negative
bacteria including most strains resistant to aminoglycosides or
3rd-generation cephalosporins. It is highly resistant to hydrolysis by
most beta-lactamases, and has low affinity for
chromosomally-encoded beta-lactamases, and exhibits rapid
penetration into gram-negative bacterial cells. In studies using
Escherichia coli and *Enterobacter cloacae*, Cefepime bound with
highest affinity to penicillin-binding protein (PBP) 3 followed by PBP 2,
then PBPs 1a and 1b. Binding to PBP 2 occurs with significantly
higher affinity than that of other parenteral cephalosporins. This may
enhance its antibacterial activity. The moderate affinity of Cefepime
for PBPs 1a and 1b probably also contributes to its overall
bactericidal activity. Cefepime has been shown to be bactericidal by
time-kill analysis (killing-curves) and by determination of minimum
bactericidal concentrations (MBC) for a wide variety of bacteria. The
cefepime MBC/MIC ratio was ≤ 2 for >80% of isolates of all
gram-positive and gram-negative species tested. Synergy with
aminoglycosides has been demonstrated *in vitro*, primarily with
Pseudomonas aeruginosa isolates.

Microbiology :

Cefepime has been shown to be active against most strains of the
following organisms:

Gram-Positive Aerobes : *Staphylococcus aureus* (including beta-
lactamase-producing strains), *Staphylococcus epidermidis* (including
b-lactamase-producing strains), other staphylococci including *S.*
hominis and *S. saprophyticus*, *Streptococcus pyogenes* (Group A
streptococci), *Streptococcus agalactiae* (Group B streptococci),
Streptococcus pneumoniae (including intermediate penicillin-resistant
strains with penicillin MIC of 0.1-1 mcg/mL), other beta-hemolytic
streptococci (Group C, G, F), *S. bovis* (Group D), *streptococcus*
viridans.

(Most strains of enterococci eg, *Enterococcus faecalis* and
methicillin-resistant staphylococci are resistant to most
cephalosporins including Cefepime).

Gram-Negative Aerobes : *Pseudomonas* sp (including *P. aeruginosa*,
P. putida, *P. stutzeri*), *Escherichia coli*, *Klebsiella* sp (including *K.*
pneumoniae, *K. oxytoca*, *K. ozaenae*), *Enterobacter* sp (including *E.*
cloacae, *E. aerogenes*, *E. agglomerans*, *E. sakazakii*), *Proteus* sp
(including *P. mirabilis*, *P. vulgaris*), *Acinetobacter calcoaceticus*
(subspecies *anitratus*, *lwoffii*), *Aeromonas hydrophila*,
Capnocytophaga and *Citrobacter* sp (including *C. diversus*, *C.*
freundii), *Campylobacter jejuni*, *Gardnerella vaginalis*, *Haemophilus*
ducreyi, *Haemophilus influenzae* (including beta-lactamase-producing
strains), *Haemophilus parainfluenzae*, *Hafnia alvei*, *Legionella* sp,
Morganella morganii, *Moraxella catarrhalis* (Branhamella catarrhalis,
including beta-lactamase-producing strains), *Neisseria gonorrhoeae*
(including beta-lactamase-producing strains), *Neisseria meningitidis*,
Providencia sp (including *P. rettgeri*, *P. stuartii*), *Salmonella* sp,
Serratia (including *S. marcescens*, *S. liquefaciens*), *Shigella* sp and
Yersinia enterocolitica.

Note: Cefepime is inactive against many strains of *Stenotrophomonas*
maltophilia (formerly known as *Xanthomonas maltophilia* and
Pseudomonas maltophilia).

Anaerobes : *Bacteroides* sp, *Clostridium perfringens*, *Fusobacterium*,
Mobiluncus, *Peptostreptococcus*, *Prevotella melaninogenica* and
Veillonella spp.

Note: Cefepime is inactive against *Bacteroides fragilis* and
Clostridium difficile.

CLINICAL PHARMACOLOGY:

Cefepime is given by injection as the hydrochloride. It is rapidly and
almost completely absorbed on intramuscular injection and mean

peak plasma concentrations of about 14 and 30 micrograms/ml have
been reported about 1.5 hours after doses of 500mg and 1g
respectively. Within 30 minutes of similar intravenous doses, peak
plasma concentrations of about 40 and 80 micrograms/ml are
achieved. The plasma half-life of Cefepime is about 2 hours and is
prolonged in patients with renal impairment. About 20% of Cefepime
is bound to plasma proteins. Cefepime is widely distributed in body
tissues and fluids. High concentrations are achieved in bile. Low
concentrations have been detected in breast milk. Cefepime is
eliminated principally by the kidneys and about 85% of a dose is
recovered unchanged in the urine. Cefepime is substantially removed
by haemodialysis.

DOSAGE AND ADMINISTRATION:

Vaxcel Cefepime can be administered either intravenous or intra
muscularly.

Adults : Usual Dosage: 1g administered IV or IM every 12 hrs. Usual
Duration of Therapy: 7-10 days, but may be longer in severe
infections. For empirical treatment of febrile neutropenia, usual
duration of therapy is 7 days or until resolution of neutropenia.
However, the dosage and route vary according to susceptibility of the
causative organism, the severity of the infection, renal function and
overall condition of the patient.

Table 1. Recommended Dosage Schedule		
Severity of Infection	Dose and Route of Administration	Dosing Interval
Mild to moderate urinary tract infections	500mg to 1g IV or IM	every 12 hrs
Mild to moderate infections other than UTI	1g IV or IM	every 12 hrs
Severe infections	2g IV	every 12 hrs
Very severe or life - threatening infections	2g IV	every 8 hrs

Paediatric (1 month up to 12 years with normal renal function) :

Usual Recommended Dosages :

Pneumonia, Urinary Tract Infections, and Skin and Skin Structure Infections :

Patients >2 months with body weight ≤ 40 kg : 50mg/kg every 12 hrs
for 10 days. For more severe infections, a dose schedule of every 8
hrs can be used.

Septicemia and Empiric Treatment of Febrile Neutropenia :

Patients >2 months with body weight ≤ 40 kg : 50mg/kg every 8 hrs
for 7-10 days. Experience of use with cefepime in paediatric patients
< 2 months of age is limited. While this experience has been attained
using 50mg/kg dose, modeling of pharmacokinetic data obtained in
patients > 2 months of age suggests that a dosage of 30mg/kg every
12 or 8 hours may be considered for patients aged 1 month up to 2
months. Administration of Vaxcel Cefepime 1g Injection in these
patients should be carefully monitored.

For paediatric patients with body weight > 40kg, adult dosing
recommendations apply. For patients > 12 years who are < 40kg, the
dosage recommendations for younger patients < 40kg should be
used. Dosage in paediatric patients should not exceed the maximum
recommended dosage in adults (2g every 8 hours). Experience with
IM administration in paediatric patients is limited.

Patients with Impaired Renal Function: The initial dose of cefepime
is the same as in patients with normal renal function. The
recommended maintenance dosages of cefepime in adult patients
with renal insufficiency as below.

Table 2. Maintenance Dosing Schedule in Patients with Renal Impairment.				
Creatinine Clearance (mL/min)	Recommended Dosing Schedule			
>50	Usual dose, no adjustment necessary			
	Very severe or life-threatening infections	Severe infection	Mild to moderate infection other than UTI	Mild to moderate urinary tract infections
	2g q 8 hrs	2g q 12 hrs	1g q 12 hrs	500mg q 12 hrs
30-50	2g q 12 hrs	2g q 24 hrs	1g q 24 hrs	500mg q 24 hrs
11-29	2g q 24 hrs	1g q 24 hrs	500mg q 24 hrs	500mg q 24 hrs
≤ 10	1g q 24 hrs	500mg q 24 hrs	250mg q 24 hrs	250mg q 24 hrs

When only steady-state serum creatinine levels are available,
following formula (Cockcroft and Gault equation) may be used to
estimate the creatinine clearance. The serum creatinine clearance
should represent a steady state of renal function.

$$\text{Males creatinine clearance : } \frac{\text{Weight (kg)} \times (140 - \text{age})}{72 \times \text{serum creatinine concentration (mg/dL)}} \text{ (mL/min)}$$

Female : 0.85 x value calculate using formula for males

Dialysis patients :

In patients undergoing hemodialysis, approximately 68% of the total
amount of cefepime present in the body at the start of the dialysis
will be removed during a 3-hr dialysis period. In patients undergoing
continuous ambulatory peritoneal dialysis, cefepime may be
administered at the same doses recommended for patients with
normal renal function, ie 500mg, 1g or 2g depending on infection
severity, at a dosage interval of every 48 hrs.

Children with impaired renal function :

Since urinary excretion is the primary route of elimination of
Cefepime in paediatric patients (see PHARMACODYNAMICS), an
adjustment of the dosage of Vaxcel Cefepime should also be
considered in patients < 12 years of age with renal impairment.
A doses of 50mg/kg in patients aged 2 months up to 12 years, and
dose of 30mg/kg in patients aged 1 month up to 2 months, are
comparable to a dose of 2 g in an adult. As recommended in table 2
above, the same increase in interval between doses and / or
reduction in dose should be used. When only serum creatinine is
available, creatinine clearance maybe estimated using either of the
following methods:

$$\text{Creatinine clearance (mL/min/1.73m}^2\text{)} : \frac{0.55 \times \text{height (centimeters)}}{\text{Serum creatinine (mg/dL)}}$$

Or

$$\text{Creatinine clearance (mL/min/1.73m}^2\text{)} : \frac{0.52 \times \text{height (centimeters)} - 3.6}{\text{Serum creatinine (mg/dL)}}$$

Impaired Hepatic Function : No adjustment is necessary.

ROUTE OF ADMINISTRATIONS:

Administration:

Intravenous (IV) : The IV administration route is preferable for
patients with severe or life-threatening infections, particularly if the
possibility of shock is present.

For direct IV administration, Vaxcel Cefepime 1g Injection should be
reconstituted with of Sterile Water for Injection, 5% Dextrose
Injection or 0.9% Sodium Chloride, using the diluent volumes
shown in Table 3.

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The resulting solution should be injected slowly inject directly into the vein over a period of 3-5 min or injected into the tubing of an administration set while the patient is receiving a compatible IV fluid (see Compatibility and Stability).

Table 3: Preparations of Solutions of Vaxcel Cefepime 1g Injection			
	Amount of diluent to be added (mL)	Approximate volume in vial (mL)	Approximate cefepime concentration (mg/mL)
IV 1g vial	10	11.4	90
IM 1g vial	3.0	4.4	230

For IV infusion, constitute the 1g vial, as noted above for direct IV administration and add an appropriate quantity of the resulting solution to an IV container with one of the compatible IV fluids (see Compatibility and Stability under Cautions for Usage).

Intramuscular (IM) : Vaxcel Cefepime 1g Injection should be constituted with one of the following diluents: using the volume shown above, Sterile Water for Injection, 0.9% Sodium Chloride Injection, 5% Dextrose Injection, or Benzyl Alcohol (see Table 3). Then administered by deep IM injection into a large muscle mass (such as the upper outer quadrant of the gluteus maximus. In pharmacokinetic study, doses up to 1g (volume < 3.1mL) were administered at single injection sites. Although Vaxcel Cefepime 1g Injection can be constituted with 1% Lidocaine HCl, it is usually not required because Vaxcel Cefepime 1g Injection causes little or no pain upon IM administration.

Compatibility and Stability :

Intravenous : Vaxcel Cefepime is compatible at concentrations between 1 and 40mg/mL with one of the following IV infusion fluids : 0.9% Sodium Chloride Injection, 5% and 10% Dextrose Injections, M/6 Sodium Lactate Injection, 5% Dextrose and 0.9% Sodium Chloride Injection, Lactated Ringers and 5% Dextrose Injection. These solutions are stable up for 7 days under refrigeration (2 – 8°C).

Vaxcel Cefepime admixture compatibility and stability information is summarized in table 4.

Incompatibility : Solutions of Vaxcel Cefepime, like those of most beta-lactam antibiotics, should not be added to solutions of metronidazole, vancomycin, gentamicin, tobramycin sulfate, or netilmicin sulfate because of physical or chemical incompatibility. However, if concurrent therapy with Vaxcel Cefepime is indicated, each of these antibiotics can be administered separately.

Intramuscular : Vaxcel Cefepime constituted as directed (in table 3) is stable for 7 days under refrigeration when using the following diluents: Sterile Water for Injection, 5% Dextrose Injection, Benzyl Alcohol, or 1% Lidocaine Hydrochloride.

NOTE: Parenteral drugs should be inspected visually for particulate matter before administration, and not used if particulate matter is present. As with other cephalosporins, the color of Vaxcel Cefepime powder and solution may darken on storage; however, product potency is unaffected.

Table 4. Cefepime Admixture Stability.				
Concentration (mg/mL)	Admixture and Concentration	IV Infusion Solutions	Stability Time for	
			RT/L (hrs)	Refrigeration
40mg/mL	amikacin 6mg/mL	NS	Freshly reconstituted	7 days
40mg/mL	ampicillin 1mg/mL	D5W	8	8 hrs
40mg/mL	ampicillin 10mg/mL	D5W	Freshly reconstituted	Freshly reconstituted
40mg/mL	ampicillin 1mg/mL	NS	Freshly reconstituted	48 hrs
40mg/mL	ampicillin 10mg/mL	NS	Freshly reconstituted	Freshly reconstituted
4mg/mL	ampicillin 40mg/mL	NS	Freshly reconstituted	Freshly reconstituted
40mg/mL	clindamycin 6 mg/mL	NS	Freshly reconstituted	7 days
4mg/mL	heparin 10 units/mL	NS	Freshly reconstituted	7 days
4mg/mL	potassium chloride 10 mEq/L	NS	Freshly reconstituted	Freshly reconstituted
4mg/mL	theophylline 0.8 mg/mL	D5W	Freshly reconstituted	Freshly reconstituted
0.2 mg/mL	NA	Peritoneal dialysis solution*	24	7 days

* "Impersol with 4.25% dextrose"; "NA = Not applicable"; "RT/L = room temperature and light"; "D5W = 5% Dextrose Injection".

CONTRAINDICATIONS:

Vaxcel Cefepime is contraindicated in patients who have had previous hypersensitivity reactions to any component of the formulation, the cephalosporin class of antibiotics penicillins or to other β -lactam antibiotics.

SIDE EFFECTS:

Vaxcel Cefepime 1g Injection is generally well tolerated. The following adverse events occurred at an incidence of > 0.1-1% (except where noted) were :
Hypersensitivity : rash, pruritus, urticaria.
Gastrointestinal : Nausea, vomiting, oral moniliasis, diarrhea and colitis (including pseudomembranous colitis).
Central Nervous System : headache.
Others : fever, vaginitis and erythema.
Events that occurred between 0.05-0.1% were abdominal pain, constipation, vasodilation, dyspnea, dizziness, paresthesia, genital pruritus, taste perversion, chills and unspecified moniliasis.
Events of clinical significance that occurred at an incidence of < 0.05% included anaphylaxis and seizure.
Local reactions at the site if IV injection occurred in 5.2% of patients this included phlebitis and Inflammation. IM administration of Vaxcel Cefepime 1g Injection was very well tolerated with 2.6% of patients experienced inflammation or pain at the injection site.
Laboratory test abnormalities : that developed during clinical trials, in patients with normal baseline value were transient. Those occurred at an incidence between 1% and 2% where elevations in alanine aminotransferase, aspartate aminotransferase, alkaline

phosphatase, total bilirubin, eosinophilia, anemia, thrombocytopenia, prolonged prothrombin time, partial thromboplastin time, and positive Coombs' test without hemolysis. Transient elevations of Blood Urea Nitrogen, and/or serum creatinine and transient thrombocytopenia were observed in 0.5-1% of patients.

Transient leukopenia, neutropenia were also seen

During post-marketing experience, agranulocytosis has been reported rarely; encephalopathy, seizures, myoclonus and / or renal failure have been reported in patients with renal impairment who received unadjusted doses of cefepime (see Precautions). Because of uncontrolled nature of these spontaneous reports, a causal relationship to cefepime has not been determined. The following adverse reactions and altered laboratory tests have been reported for cephalosporins-class antibiotics Stevens-Johnson syndrome, erythema multiforme, toxic epidermal necrolysis, toxic nephropathy, aplastic anemia, hemolytic anemia, hemorrhage and false-positive tests for urinary glucose. The safety profile of Vaxcel Cefepime in infants and children are similar to that seen in adults. The most frequently-reported adverse event considered related to Vaxcel Cefepime in clinical trial was rash.

SYMPTOMS AND TREATMENTS FOR OVERDOSAGE:

In case of severe overdosage, especially in patients with compromised renal function, dialysis will aid in the removal of Cefepime from the body; peritoneal dialysis is of no value. Accidental overdosing can occur if large doses are given to patients with reduced renal function (see Precautions).

PRECAUTIONS:

Serious and occasionally fatal hypersensitivity reactions (including anaphylactoid and severe cutaneous reactions) have been reported in patients receiving therapy with beta-lactams. Before initiating therapy with Vaxcel Cefepime-1g Injection, careful inquiry should be made concerning previous hypersensitivity reactions to penicillins, cephalosporins, carbapenems or other beta-lactam agents. If an allergic reaction occurs, Vaxcel Cefepime-1g Injection must be discontinued immediately and appropriate alternative therapy instituted.

Antibiotics should be administration with caution is required in any patient who has demonstrated some form of allergy, particularly to drugs. If an allergic reaction to Vaxcel Cefepime 1g Injection occurs, discontinue the drug and treat the patients appropriately. Serious immediate hypersensitivity reactions may require epinephrine and other supportive therapy.

Pseudomembranous colitis has been reported with most broad-spectrum antibiotics, including cefepime. It is important to consider this diagnosis in patients who develop diarrhea in association with the use of antibiotics. Mild cases of colitis may respond to drug discontinuation alone, moderate to severe cases may require more elaborate management.

As with other antibiotics, use of Vaxcel Cefepime 1g Injection may result in overgrowth of non susceptible organisms. Should superinfection occur during therapy, appropriate measures should be taken.

In patients with impaired renal function, such as reduction of urinary output because of renal insufficiency (creatinine clearance $\leq$ 50 mL/min) or other conditions that may compromise renal function, the dosage of Vaxcel Cefepime 1g Injection should be adjusted to compensate the slower rate of renal elimination (see Dosage and Administration and Clinical Pharmacology). Serious adverse events, including encephalopathy, seizures, myoclonus and / or renal failure have been reported in postmarketing experience in patients with renal impairment who received unadjusted dosage of cefepime injection. Renal function should be monitored carefully if drugs with

nephrotoxic potential such as aminoglycosides and potent diuretics are administered with Vaxcel Cefepime 1g Injection.

Carcinogenicity, Mutagenicity & Impairment of Fertility : No long-term studies in animals have been performed to evaluate carcinogenic potential. *In vitro* and *in vivo* tests for genotoxicity have shown that cefepime is not genotoxic. No impairment of fertility has been seen in rats.

Use in pregnancy & lactation : Reproductive studies in mice, rats and rabbits showed no evidence of fetal damage; however, there are no adequate and well-controlled studies in pregnant women. Because animal reproduction studies are not always predictive of human response, Cefepime should be used during pregnancy only if clearly needed.

Nursing Mothers : Cefepime is excreted in human breast milk in very low concentrations. Caution should be used when cefepime is administered to a nursing woman.

Geriatric Use : In clinical studies, when geriatric patients received the usual recommended adult dose, clinical efficacy and safety were comparable to clinical efficacy and safety in non geriatric adult patients. There was a modest prolongation in elimination half-life and lower renal clearance values compared to those seen in younger persons. Dosage adjustment is recommended if renal function is compromised (see Dosage and Administration).

DRUG INTERACTIONS:

The pharmacokinetics of cefepime and amikacin when given concurrently to patients with normal renal function do not appear to be altered. Concomitant administration of probenecid prolongs the renal tubular secretion of cephalosporins resulting in high serum concentrations of these antibiotics.

STORAGE:

Store below 30°C. Protect from light.
For storage after reconstitution please refer section Route of Administration.

**KEEP OUT OF REACH OF CHILDREN
JAUHI DARI KANAK-KANAK**

PACK QUANTITIES:

Single vial per box and 10 x 1g per box.

Further information can be obtained from pharmacist, physician or the manufacturer.

Product Registration Holder / Manufacturer:

 **Kotra Pharma (M) Sdn. Bhd.**
No.1, 2 & 3, Jalan TTC 12, (90082-V)
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